

Lecture 10

Cantor Sets

- $C = \bigcap_{n=1}^{\infty} C^n$
- $C^n = 2^n$ subsets of length $\frac{1}{3}$
- $C^1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$
- $C^2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$
- C is the middle thirds Cantor set, obtained by removing middle thirds of remaining intervals
- 4 major (topological) properties:
 - Nonempty
 - Consider $E^n =$ set of end points of the C^n intervals
 - $E = \bigcup E^n$ is countable, contained in C
 - E is dense in C
 - C is uncountable
 - Compact
 - Obvious to show
 - Perfect
 - Endpoints are close to intervals
 - Totally disconnected
 - Many small clopen subsets

Moore-Klein Theorem: M is a Cantor space iff $M \cong C$

- M is a Cantor space if it has the 4 properties
- C refers to the middle thirds Cantor set
- A Cantor piece is a clopen of subset of M with the 4 properties, which is another Cantor space

Words

- $W(n) = \{\text{words in 2 letters } (0, 2) \text{ of length } n\}$
- $W(\infty) =$ all 2 letter words (the limit, rather than the union)
- Using words to label Cantor pieces

2-part Lemma

- Can divide M into finitely many ε small disjoint Cantor pieces
- Can divide any Cantor piece into 2 Cantor pieces
- 2-parts imply that dyadically many intervals can be formed, all of length less than some ε
 - Part 1: Say we have some number of Cantor pieces, all of which can be covered by subsets of $\text{diam} < \varepsilon$
 - Part 2: We can split one part at a time until we arrive a dyadic number, each split results in two smaller Cantor pieces

Proof

- C^n and $W(n)$ have the same cardinality so there exists a bijection
- Suppose M has a clopen disjoint covering by Cantor pieces of diameter $< \varepsilon_1$
- $\mathcal{M}_1 =$ the members of the disjoint covering using the ε_1 sized pieces, with a total of 2^{k_1} pieces
 - Can be represented as $W(k_1)$
- $K \in \mathcal{M}_1$ can be split into another collection of members of a disjoint covering of itself, with a total of 2^{k_2} pieces
 - Can be represented as $W(k_2)$

- Let the union of all collections members of disjoint coverings of members of the disjoint covering of M be called $\mathcal{M}_2$
- However, there are 2^{k_1} possible $K \in \mathcal{M}_1$, so we use $W(k_1) \times W(k_2)$ to represent the elements of $\mathcal{M}_2$
- We can keep repeating this process into an infinite sequence of words, which labels the points of M
- Then, there exists a bijection from $W(\infty) \rightarrow M$
- There also exists a bijection from $C \rightarrow W(\infty)$, so there exists a bijection from $C \rightarrow M$
 - Will this be continuous? Yes!
 - For 2 close points in C , the words will have the same prefix
 - Same prefix implies being close
 - Both C and M are compact, so there is a homeo since there exists a continuous bijection from C to M

Outer Measure

- The outer measure of a set $S \subset \mathbb{R}$ is the infimum of sums of length $\sum_{i=1}^{\infty} I_i$ where $\cup I_i \supset S$
 - Each I_i is an interval (a_i, b_i) , of length $b_i - a_i$
 - Ex. $\mathbb{Q} \cap [0, 1]$ has measure 0
- Middle thirds Cantor set has measure 0
- Not all middle thirds-type setups produce sets of measure 0
- Take small amount away from the middle, and even less from the remaining
 - End up with positive measure
 - Still has 4 properties of Cantor set
- Can also have non-fractal structure, and still is a Cantor space
 - Take away intervals of different length and amounts
 - Make sure to not leave any intervals untouched through iterations
- Cantor set can't have measure 1
- Sum of measure of Cantor set and the complement is 1

Cantor Surjection Theorem: There exists a continuous surjection from C to any nonempty, compact metric space M .

- Can be used to digitize data sets